

Cyber Resilience

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What Is Crisis Leadership?

- Crisis leadership = Leading under uncertainty, time pressure, and incomplete information.
- Key characteristics:
 - Fast decision-making
 - High accountability
 - Emotional control
 - Clear communication
 - Ethical judgment
- “Would you prefer a technically skilled leader who panics, or a calm leader who decides quickly?”

Why Crisis Leadership Fails

- Common causes:
 - Panic
 - Lack of authority
 - No decision hierarchy
 - Internal conflicts
 - Delayed communication
- Governance must support leadership.



Leadership vs Management in Crisis

- During normal times: Management dominates.
- During crisis:
 - Uncertainty increases.
 - Time pressure increases.
 - Fear increases.
- Planning becomes less useful. Decisiveness becomes critical.
- In crisis, leadership dominates.
- Management keeps the organization running. Leadership keeps the organization together.

Management

Leadership

Processes

Direction

Structure

Vision

Control

Inspiration

Planning

Decisiveness

Role of CEO During Crisis

- CEO must:
 - Take visible control
 - Approve major decisions
 - Communicate externally
 - Balance legal & reputational risk
 - Protect long-term strategy
- Important:
 - Silence = loss of trust.

Decision-Making Under Pressure

- Characteristics:
 - Limited information
 - High uncertainty
 - Time constraint
 - Public visibility
- Perfection is impossible.
- Speed matters.

Simulation: “You Have 5 Minutes”

- Scenario: You are the CEO of a medium-sized fintech company.
 - 09:10 AM – IT reports:
 - Suspicious activity in payment system
 - Possible unauthorized transactions
 - Cause unknown
 - Media not yet aware
 - Customers are currently making payments.
- You Have 5 Minutes to Decide
 - Decision 1: System Status
 - Keep platform fully running
 - Partially restrict payment features
 - Shut down entire payment system
 - Decision 2: Communication
 - Wait until investigation is complete
 - Send internal message only
 - Issue short public holding statement
 - Decision 3: Escalation
 - Handle internally
 - Activate full crisis team
 - Inform board immediately

Escalation Twist (After 5 Minutes)

- New update:
 - Fraud confirmed.
 - 200 customers affected.
 - Journalist calls for comment.
- Do you change your decision?
- What do you communicate?
- Who speaks publicly?

The 4-Step Crisis Decision Model

- Assess impact
- Prioritize business continuity
- Decide & act
- Communicate immediately

Delay increases escalation.

DELAY → CONFUSION →
ESCALATION →
REPUTATIONAL DAMAGE



Ethical Dilemmas in Crisis

- Examples:
 - Pay ransomware or not?
 - Disclose breach immediately?
 - Shut down services to prevent spread?
- Crisis leadership involves ethics.

Ethical Dilemma Debate: “To Pay or Not to Pay?”

- Scenario: Your company has been hit by ransomware.
 - All systems encrypted.
 - Backups partially affected.
 - Downtime costs €250,000 per day.
 - Attackers demand €1 million.
 - They claim customer data will be leaked if you refuse.
 - Regulators are not yet aware.
 - Media does not know.
- You are the executive team.
- **Should the company pay the ransom?**
- Divide into 3 groups:
 - Group A – Pay the Ransom
 - Group B – Do NOT Pay
 - Group C – Board / Ethics Committee (judges)

Crisis Communication Principles

- Crisis Communication Principles:
 - Be fast
 - Be transparent
 - Be consistent
 - Avoid speculation
- Reputation risk is often larger than operational damage.

Bad Communication

“Minor technical issues.”
(Minimizes problem)

Avoids confirming fraud

No action explained

No accountability shown

No timeline for update

Defensive tone

Creates suspicion if facts
emerge

Increases reputational risk

Good Communication

Clearly acknowledges
unauthorized transactions

Confirms what is known

Explains immediate actions taken

Shows responsibility and control

Sets clear update timeframe

Calm, professional tone

Builds trust through transparency

Protects reputation

Stakeholder Mapping

- During crisis, leadership must identify and prioritize stakeholders:
 - Employees
 - Customers
 - Regulators
 - Media
 - Suppliers
 - Shareholders
- Poor communication = escalation.



Sector Differences

- Same BCM principles → different priorities.
- BCM protects what matters most — and what matters most depends on the sector.

Sector

Priority

Hospital

Patient safety

E-commerce

Revenue & uptime

Municipality

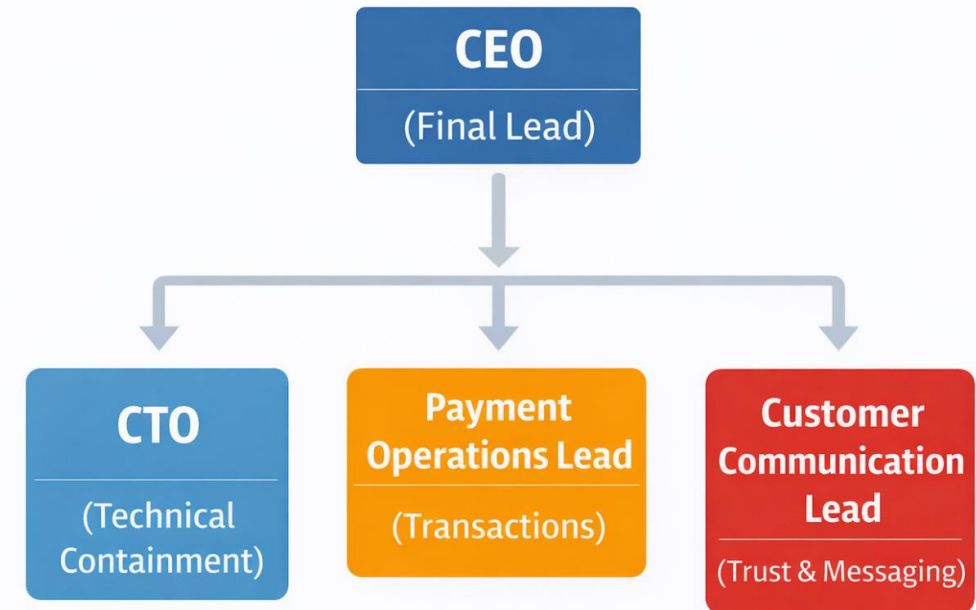
Public service
continuity

Hospital Governance Example

- Key crisis leaders:
 - Hospital director
 - Chief medical officer
 - IT security officer
 - Communications officer
- Priority:
 - Human life > IT systems > Reputation

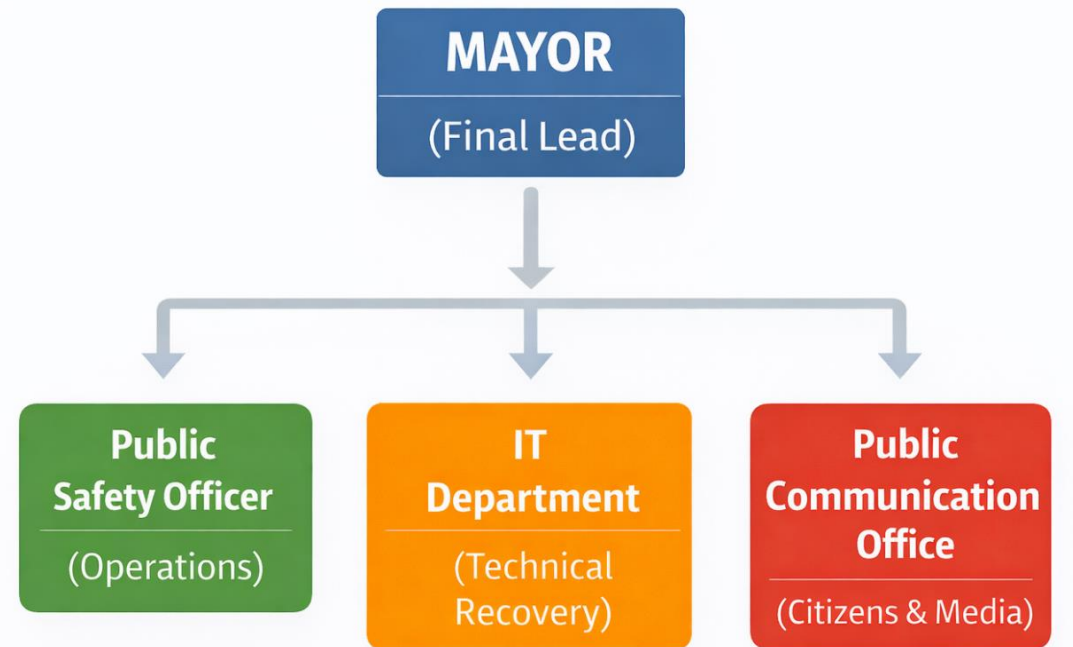
E-Commerce Governance Example

- Crisis team:
 - CEO
 - CTO
 - Payment operations lead
 - Customer communication lead
- Priority:
 - Revenue continuity & customer trust



Municipality Governance Example

- Crisis team:
 - Mayor
 - Public safety officer
 - IT department
 - Public communication office
- Priority:
 - Public safety & legal compliance



Crisis Governance Comparison

Element	 Hospital	 E-Commerce	 Municipality
Primary Objective	Protect human life	Maintain revenue flow	Protect public safety
Top Priority	Patient safety	Revenue continuity & uptime	Public safety & legal compliance
Crisis Leader (Final Authority)	Hospital Director	CEO	Mayor
Operational Lead	Chief Medical Officer (CMO)	CTO	Public Safety Officer
Technical Lead	IT Security Officer	CTO / IT Security	IT Department
Communication Lead	Communications Officer	Customer Communication Lead	Public Communication Office
Main Risk in Downtime	Risk to patient lives	Financial loss & customer churn	Disruption of public services
Ethical Sensitivity	Very high (life & health)	Medium (financial impact)	High (citizen impact & law)
Reputation Impact	Trust in healthcare	Brand trust & loyalty	Political & public trust



Exercise

- Design BCM Governance Structure for:
 - Group 1 → Hospital
 - Group 2 → E-commerce company
 - Group 3 → Municipality
- Include:
 - Who owns BCM?
 - Who leads crisis?
 - Who communicates?
 - Who is backup decision-maker?

Answer:

BCM GOVERNANCE COMPARISON		
		
HOSPITAL	E-COMMERCE	MUNICIPALITY
BCM OWNER	BCM OWNER	BCM OWNER
 Hospital Director	CEO	Mayor
 Director + CMO	CEO (+ CTO tech)	Mayor
 Comm. Officer + Medical rep	Customer Comm. Lead	Public Comm. Office
 Deputy Director	COO/Deputy CEO	Deputy Mayor
TOP PRIORITY	REVENUE & TRUST	TOP PRIORITY
Human Life	Revenue & Trust	Public Safety
Hospital → Ethics & Life First → E-Commerce → Revenue & Trust First → Municipality → Public Safety		
Hospital → Ethics & Life First → E-Commerce → Revenue & Trust First → Municipality → Public Safety & Law First		

Key Takeaways

- Scope defines boundaries
- Policy defines commitment
- Governance defines authority
- Management owns BCM
- Structure prevents chaos
- <https://www.youtube.com/watch?v=yBL4eco0NCs>
- https://www.youtube.com/watch?v=bHs_VQZ1xfs

BCMS Fundamentals

- A Business Continuity Management System (BCMS) is a structured framework that helps an organization prepare for, respond to, and recover from disruptions while maintaining critical business operations.
- Key idea: Not only reacting to crises — but planning continuity in advance.
- Typical disruptions:
 - Cyberattacks
 - System failures
 - Natural disasters
 - Supplier outages
 - Human errors
- Goal:
 - Maintain critical business processes during disruptions.
 - Standard reference:
 - ISO 22301 – Business Continuity Management.

Alignment with Strategy

- BCMS must support the **organization's strategic objectives.**
- BCMS protects **what is most important for the organization.**

Business Strategy

E-commerce company →
24/7 availability

Hospital → patient safety

Municipality → public
services

BCMS Alignment

Ensure webshop uptime

Ensure medical systems
always available

Ensure emergency services
remain operational

The Concept of a Business Process

- A **business process** is a **set of activities that produce a service or product for customers or stakeholders to achieve a business goal.**
- Examples:
 - Order processing
 - Payroll management
 - Patient treatment
 - Online payment processing
- Characteristics:
 - Has inputs and outputs
 - Involves people and systems
 - Supports organizational goals
- In BCMS:
 - **Processes are the starting point for continuity planning.**

Process Dependencies

- **Dependencies** are the **resources that a business process needs in order to function**. Every business process depends on **resources**.
- Example of Business process: **Online order processing**
- Dependencies:
 - Webshop system
 - Payment provider
 - Inventory database
 - Warehouse staff
 - Internet connectivity
- If one dependency fails → the process may stop.

Dependency	Example
IT systems	ERP, databases, cloud platforms
People	Employees, specialists
Infrastructure	Internet, power
Suppliers	Payment providers, logistics

A visual diagram

- Process → Dependencies → Disruption → Impact (BCMS Concept)
- **A business process only works if its dependencies work.**
- **If a dependency fails → disruption occurs → business impact appears.**
- This is why BCMS focuses on:
 - identifying **critical processes**
 - mapping **dependencies**
 - preparing **recovery plans**

BUSINESS PROCESS
(Order Processing)



DEPENDENCIES

- IT systems (Webshop, Database)
- People (Customer support)
- Infrastructure (Internet, Power)
- Suppliers (Payment provider)



DISRUPTION

- Cyberattack
- System outage
- Power failure
- Supplier failure



IMPACT

- Orders cannot be processed
- Revenue loss
- Customer dissatisfaction
- Reputational damage

Exercise

- Process Mapping Exercise
- Work in **small groups (3–4 students)**.
- Task:
 - Identify **three business processes** in the following example organization:
- Example organizations:
 - Hospital
 - Webshop
 - Municipality
 - Bank
- For each process identify **dependencies**.

Example solution structure:

- **Processes rely on multiple components.**
- BCMS focuses on protecting processes and their dependencies, because if dependencies fail, the process stops.
- A process only works if its dependencies (people, IT, suppliers, facilities) are available.

Business Process	IT	People	Suppliers
Online order processing	Webshop system	Customer support	Payment provider
Payroll processing	HR software	HR staff	Payroll provider
Inventory management	ERP system	Warehouse staff	Logistics partner

Answer

- In BCMS, disruptions often occur because dependencies fail, not the process itself.

Organization	Business Process	Key Dependencies
Hospital	Patient treatment	Doctors, nurses, medical equipment, patient records system
Hospital	Emergency services	Ambulances, emergency staff, communication systems
Hospital	Medication management	Pharmacy staff, drug inventory system, suppliers
Webshop	Order processing	Website platform, database, payment system
Webshop	Online payment processing	Payment gateway, banking network, fraud monitoring
Webshop	Product delivery	Warehouse staff, logistics providers, inventory system
Municipality	Citizen registration	Government database, civil servants, IT systems
Municipality	Emergency response	Police/fire departments, communication systems
Municipality	Permit processing	Administrative staff, document management system
Bank	Account management	Banking software, databases, staff
Bank	Payment transactions	Payment network, IT systems, fraud detection
Bank	Loan processing	Credit systems, financial analysts, compliance systems

Exercise

- Your company
 - (2 sites)
 - Make a business impact assessment
 - Processes
 - Dependency analytics (People, premises, ICT, suppliers,...)
 - Make a risk analysis
 - 3 risks

Answer

- BCMS identifies **critical processes, their dependencies, and major risks** to ensure continuity during disruptions.
- BCMS analyzes **processes → dependencies → risks → impact** to maintain business continuity.

Element	Example	Dependencies / Cause	Impact
Business Process	Order processing	Website platform, database, IT staff	Orders cannot be processed
Business Process	Customer support	Support staff, CRM system, communication tools	Customers cannot receive assistance
Business Process	Product delivery	Warehouse, logistics provider, inventory system	Orders cannot be delivered
Dependency	People	IT administrators, support staff, warehouse employees	Operations slow or stop
Dependency	Premises	Office site, warehouse	Staff cannot access workplace
Dependency	ICT	Website, servers, internet connection	Systems unavailable
Dependency	Suppliers	Payment provider, logistics company	Payments or deliveries fail
Risk 1	Cyberattack	Weak cybersecurity controls	Website outage, possible data breach
Risk 2	Power outage	Infrastructure failure at main site	Systems and operations stop
Risk 3	Supplier disruption	Logistics company failure	Product delivery delays

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Independent Work

- Choose **one organization**:
- Examples:
 - Small webshop
 - Restaurant
 - Doctor clinic
 - Local government office
- Task:
 - Identify **5 business processes**
 - Determine which are **critical** and **non-critical**

Business Process	Critical?	Reason
Online payment	Critical	Revenue stops without it
Marketing newsletter	Non-critical	Business can operate without it
Customer support	Critical	Customer issues unresolved
Social media posting	Non-critical	No direct operational impact

Independent Work

- Assignment:
- Choose a company and:
 - Define BCM scope
 - Draft a short BCM policy outline (bullet format)
 - Draw simple governance structure
 - Identify one governance weakness
 - Create one-page BCM Policy Statement. Must include:
 - Purpose
 - Scope
 - Commitment
 - Governance responsibility
 - Review frequency